



Engineering Insights

Edition #1 — 2026-03

TL;DR

- The team advanced its key technical initiatives this month.

This Month in Engineering

Design Note: Caching Strategy for the Read Path We evaluated three caching approaches for the product catalog read path: cache-aside, write-through, and write-behind. [\[c1\]](#)

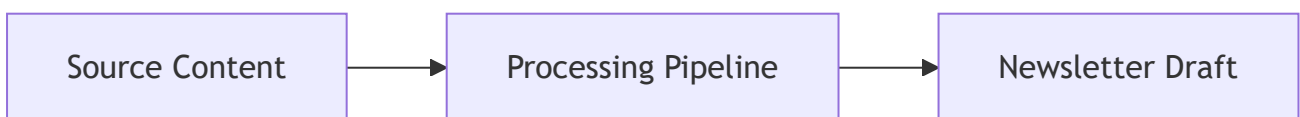
The gateway rollout needs a canary stage: route five percent of traffic through the new API gateway for 48 hours, watch the p99 latency dashboard, then ramp to fifty percent.. [\[c2\]](#)

Platform Architecture Review — May 2026 Engineering monthly sync Platform & Data teams. [\[c3\]](#)

Current State: Legacy Reverse Proxy Authentication duplicated in every microservice No centralized rate limiting Manual TLS certificate rotation Routing rules spread across twelve config files. [\[c4\]](#)

Target State: Managed API Gateway Central JWT validation at the edge Declarative routing with per-service policies Built-in rate limiting and request quotas Automatic certificate management. [\[c5\]](#)

Caching Layer Results Redis write-through cache in staging Median read latency: 120 ms to 18 ms Cache hit rate: 94 percent Next: production rollout behind a feature flag Notes: Mention that the feature flag rollout starts the first week of June.. [\[c6\]](#)



High-level flow from source content to the newsletter draft.

Action Items

- **Solution Architect:** follow up on the open design decisions.

References

This Month in Engineering

- [\[c1\]](#) 1d82e7c4-bac7-42ba-a51a-dd76e4011dd0 — caching-design-note.pdf {"page": 1}
- [\[c2\]](#) 983af916-f200-4b9f-a532-5710f339facc — deployment-notes.txt {"lines": "3-4"}
- [\[c3\]](#) ff532f7a-629c-4ae6-8dbf-3d0d677b5c3b — platform-deck.pptx {"slide": 1}
- [\[c4\]](#) 7eff3532-b993-40ce-8e43-6f5aa68c3340 — platform-deck.pptx {"slide": 2}
- [\[c5\]](#) e1813b55-0d2b-4cf7-99d1-89456fd53e53 — platform-deck.pptx {"slide": 3}
- [\[c6\]](#) 7db02738-13fd-46f7-b214-b684aa36996c — platform-deck.pptx {"slide": 4}